

Polynomial identities



1

a $a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$

b $a^2c^2 + b^2d^2 + b^2c^2 + a^2d^2$

2 No solution provided.

3 1 004 004

4 No solution provided.

5 4080

6 15689

7

a $(u + v)(u - v)$

b 5400

c 384

8 No solution provided.

9 No solution provided.

10 9 010 000

Applications

11

a 136

b Example answer:
$$x^2 \times y^2 = x^2y^2$$
$$= (xy)^2$$

12

a 16

b 49

c t^2

d 4900

e No

13

a $10m + n$



b $10n + m$

c $(10m + n) - (10n + m) = 9(m - n)$

14

a 15, 16, 17

b C and D

15

a $m^4 + 2m^2n^2 + n^4$

b No solution provided.

c Pythagora's Theorem

d 16, 34

16 100, 28, 96

17

a $x^3 - a^3$

b $97 \times 10\,309$

18

a $2x^4 + 5x^3 + 8x^2 + 6x + 4$

b 25864

c 10

19

a $x^2 = (x - 1)^2 + x - 1 + x$

b No solution provided.

c 10201

20

a 121

b 11

$$\begin{aligned}
 \text{c } x(x+1)(x+2)(x+3) + 1 &= (x^2 + x)(x^2 + 5x + 6) + 1 \\
 &= x^4 + 6x^3 + 11x^2 + 6x + 1 \\
 &= (x^2 + 3x + 1)(x^2 + 3x + 1) \\
 &= (x^2 + 3x + 1)^2
 \end{aligned}$$